

Firstname

Lastname

Group

Electronic Circuits
Second Year
Semester 4
2023/2024

Electronic Circuits Final Exam

Second year

Monday, June 3, 2024 - 9am to 10:30am

Course qst	/6
Exercise 1	/5
Exercise 2	/4
Exercise 3	/5

This is a **closed book exam**. All electronic devices are not permitted. **Do not** take pictures of the exam.

The proctors **cannot answer questions** during the exam. If you find an exam question unclear, then make reasonable assumptions, write down your assumptions, and answer the question.

Please write neatly. Write your answer using a pen or a pencil, but do not write in red or any color

that is close to red. Use your rough sheet before answering.

There are a total of **20 points**. You have **90 minutes** to write the exam.

Do not attempt to cheat, copy, talk to a classmate, or give anything to or take anything from another classmate (borrowing a correction-pen may be allowed), otherwise you may be **disqualified**.

Write your **full name** and **group** on the head of each of your answer sheets, and leave your student ID card out to be checked.

Course questions (6 marks)

Warning: Correct answer **+0.5 pt**. Wrong answer **-0.25 pt**. A mix of both or no answer **0 pt**.

1. Arduino IDE consists of a set of functions. What are they?

- ☐ loop() and build() and setup()
- ☐ build() and loop()
- ☐ setup() and build()
- ☐ setup() and loop()

2. Arduino Codes are referred to as _____ in the Arduino IDE

- ☐ sketches
- ☐ drawings
- ☐ links
- ☐ notes

3. Is the Arduino code an Object-Oriented programming language or a Procedural programming language?

- ☐ The Arduino Code follows the Top-Down Procedural ideology
- ☐ The Arduino Code follows a custom Procedural Ideology
- ☐ The Arduino Code follows the Object-Oriented ideology
- ☐ The Arduino Code follows the Bottom-Up Procedural ideology

4. What is the difference between an IDE and a compiler?

- ☐ The compiler executes the code while the IDE gives a graphical environment for writing the code
- ☐ The compiler links the code to the respective files and the IDE takes it from there

- ☐ The compiler and the IDE are the same thing
- ☐ The IDE executes the code while the compiler gives a graphical environment for writing the code

5. What will be the output of the following Arduino code?

```
#define X 10;
void setup(){
    X=0;
    Serial.begin(9600);
    Serial.print(X);
}
void loop(){
    //Do nothing...
}
```

- ☐ 0xAB
- ☐ 0xa
- ☐ 0
- ☐ Error

6. What type of signal does the analogWrite() function output?

- ☐ Pulse Code Modulated Signal
- ☐ Frequency Modulated Signal
- ☐ Pulse Width Modulated Signal
- ☐ Pulse Amplitude Modulated Signal

7. Which Raspberry Pi model has a built-in Gigabit Ethernet port?

- ☐ Raspberry Pi 2 Model B
- ☐ Raspberry Pi 3 Model B
- ☐ Raspberry Pi 4 Model B
- ☐ All of the above

8. Which Raspberry Pi model has a built-in Bluetooth module?

- ☐ Raspberry Pi 2 Model B
- ☐ Raspberry Pi 3 Model B
- ☐ Raspberry Pi 4 Model B
- ☐ Raspberry Pi Zero

9. Which Raspberry Pi model is the smallest?

- ☐ Nano
- ☐ Pico
- ☐ Zero
- ☐ None of the above

10. What is the purpose of the Raspberry Pi official power over Ethernet (PoE) hat?

- ☐ To provide power to the Raspberry Pi
- ☐ To control external devices and sensors
- ☐ To display images and interact with the Raspberry Pi
- ☐ To connect to the internet wirelessly using PoE

11. What IoT collects?

- ☐ Device data
- ☐ Machine generated data
- ☐ Sensor data

- ☐ Human generated data

12. Which of the following is false about the IoT components?

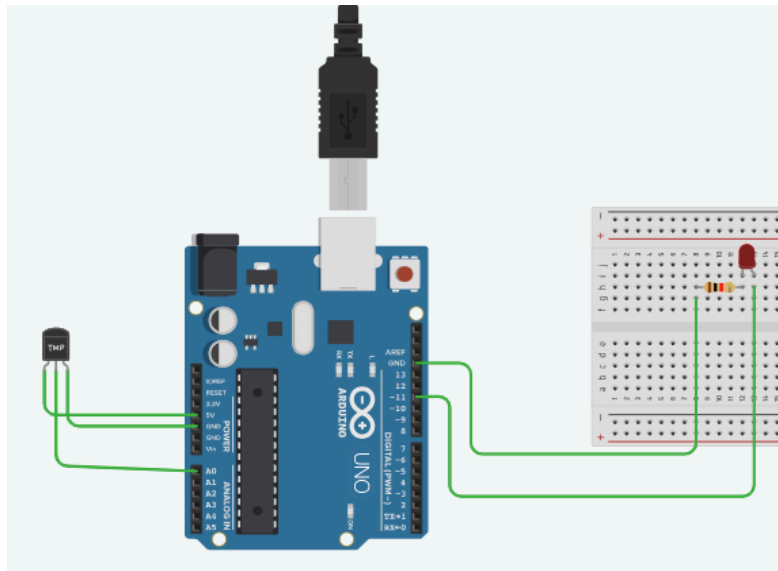
- ☐ A light sensor (photoresistor) is an analog sensor
☐ A microphone is a digital sensor
☐ A push button is a digital sensor
☐ A keyboard is a digital sensor

Exercise 1 (5 marks)

Given the circuit and the associated source code below, where we have a temperature sensor connected to an Arduino Uno that reads analog values from this sensor. We will read a value (from the pin connected to the sensor) once and control the brightness of an LED based on the read value (the brightness value will be increased/decreased by 5). The source code contains errors and is not complete.

Correct and complete the source code so that it meets the announcement of the exercise.

N.B. `analogRead(analog_pin)`: returns values in [0-1023].



```
1  int led_pin = 5;
2  int analog_pin = A0;
3  int led_brightness, read_value;
4  bool increasing = true;
5  void setup() {
6      .....
7      Serial.begin(9600);
8      read_value = analogRead(analog_pin);
9      Serial.print(read-value);
10     if (.....)
11     {
12         led_brightness = read_value / 255;
13     }
14     else
15     {
16         led_brightness = read_value;
17     }
```

```

18 .....Write(.....);
19 }
20 void loop(){
21   if( ..... >= 255)
22   {
23     increasing = false;
24   }
25   else if( ..... < 0)
26   {
27     increasing = true;
28   }
29   if(increasing == false){
30     .....; // increase the brightness
31   }
32   else {
33     .....; // decrease the brightness
34   }
35   digitalWrite(analog_pin, brightness);
36   delay(10);
37 }

```

Line	Correction

Exercise 2 (4 marks)

Let's look at the following source code. Its purpose is to vary the speed of a fan linked to Raspberry Pi Model 4:

```
1 import GPIO.RPi as GPIO
2 import time.RPi
3
4 PWM_PIN = 31
5 FREQUENCY = 25000
6
7 GPIO.setmode(RPi.BCM)
8 GPIO.setup(PWM_PIN, GPIO.IN)
9
10 pwm = GPIO.PWM(PWM_PIN, FREQUENCY)
11 pwm.start(0)
12
13 def set_fan_speed(speed):
14     if 0 <= speed <= 100
15         pwm.ChangeDutyCycle(speed)
16     else:
17         print("Speed should be between 0 and 100")
18
19
20 while True:
21     for speed in range(101, 201, 10):
22         set_fan_speed(speed*10)
23         print(f"Fan speed set to {speed}%")
24         time.wait(2)
25
26     for speed in range(201, 101, -10):
27         set_fan_speed(speed)
28         print(f"Fan speed set to {speed}%")
29         time.halt(2)
30 pwm.stop()
```

1- Using this source code, the fan won't turn. Help us fix it by filling in the table below:

Line	Correction

[illegible]

2- Update the source code in order to handle user interruption (please indicate in which line the source code should start)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

Exercise 3 (5 marks)

Here is a python source code that retrieves data from a gas sensor (MQ-135) and, depending on a threshold value, the DC motor will turn, its speed increasing according to the sensor value.

1. Complete the source code below to make it functional,

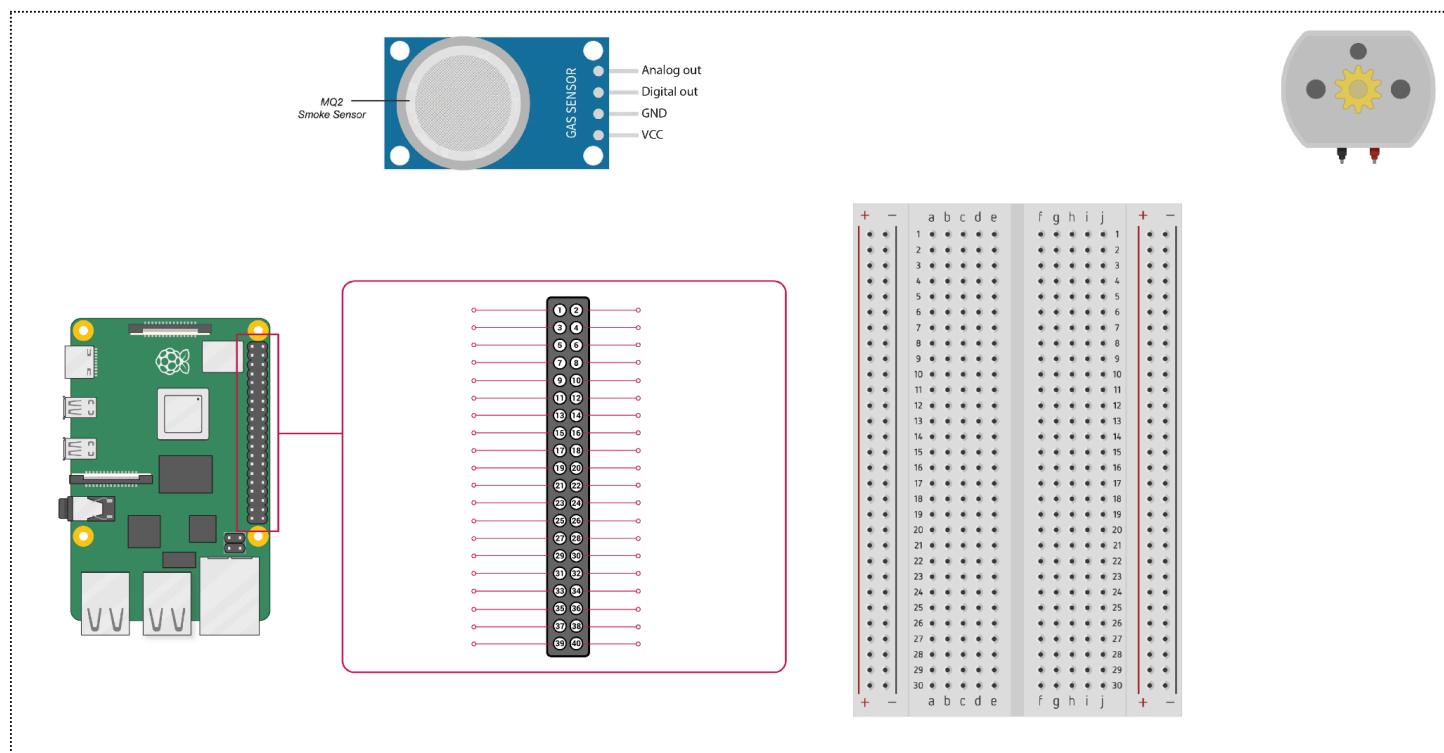
```

1 .....
2 .....
3 .....
4 .....
5 THRESHOLD = 0.5
6 .....
7 GPIO.setup(.....)
8 GPIO.setup(.....)
9 pwm = GPIO.PWM(MOTOR_PIN, 1000)
10 pwm.start(0)
11 def read_sensor(pin):
12     GPIO.setup(pin, GPIO.OUT)
13     GPIO.output(pin, GPIO.LOW)
14     time.sleep(0.1)
15     .....
16     count = 0
17     while GPIO.input(pin) == GPIO.LOW:
18         count += 1
19     return count
20
21 def map_value(value, in_min, in_max, out_min, out_max):
22     return (value - in_min) * (out_max - out_min) / (in_max - in_min) +
out_min
23
24
25 while True:
26     sensor_value = read_sensor(SENSOR_PIN)
27     voltage = sensor_value / 1000.0
28     if voltage > .....:
29         duty_cycle = map_value(voltage, THRESHOLD, 3.3, 0, 100)
30         duty_cycle = min(....., 100)
31     else:
32         duty_cycle = 0
33     pwm.ChangeDutyCycle(duty_cycle)
34     print(f"Sensor Value: {sensor_value}, Voltage: {voltage:.2f}V, Duty
Cycle: {duty_cycle}%")
35     time.sleep(1)

```

2. Complete the drawing in the figure below so that the circuit is operational. In this circuit, resistors and transistors are ignored.

Please draw clearly. If the drawing is not clear, the answer will be canceled.



You can use this part for additional source code: